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Palaeomagnetic and Rock Magnetic Studies on Cenozoic Basalts from Kyushu, Japan

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Summary

Some of these Cenozoic basalts are magnetized in either the normal or reversed direction of the axial dipole field but some others are magnetized obliquely. The remanences of the latter are quite as stable as the normal or reversed flows and are interpreted as recording geomagnetic field directions during polarity transitions. When combined with measurements made on rocks from sites some 10 km away from ours, two paths connecting these transitional virtual poles may be traced, one for the lower Pliocene and the other for the upper Pliocene. Both cross central Russia and the Indian Ocean. Palaeosecular variations of the geomagnetic field have been investigated by restricting our studies to flows which yield virtual poles at latitudes higher than 45° and the dispersion of cleaned R.M. directions obtained from such flows agree with world-wide average values. Thermomagnetic, ore microscope and electron microprobe investigations have been made and the shape of the thermomagnetic curves is related to evidence of both low and high temperature oxidation.

1. Introduction

The volcanic rocks of Kyushu Island, Japan range in age from Miocene to Pleistocene (Table 1). Palaeomagnetic studies on them have previously been reported by Kawai *et al.* (1961, 1962) and also by Nomura (1967). The latter discovered some lava flows whose virtual geomagnetic poles computed after A.F. and thermal cleaning of the N.R.M., are situated in low latitudes and suggested that they record geomagnetic field directions during polarity transitions.

Our rock samples (see Table 1) come from an area situated about 10 km E and SE of that from which Nomura collected his. The flows are labelled with letters and numbers following the system adopted by Japanese geologists who have worked in the area and also by Nomura (1967) in his palaeomagnetic study.

2. Palaeomagnetic measurements

2.1 *Method*

Ninety-three hand samples from twenty sites were collected for us from the Kishima District (33° N, 130° E) in the north-west of Kyushu Island by Dr S. Kume and Dr H. Domen of the Universities of Osaka and Yamagushi respectively. Sample

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† Received in original form 1969 August 18.

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Table 1

Geological and magnetic stratigraphy

Age	Formation	Flow No.†	Site No.	Sample Nos	Type of site	Polarity of R.M.*
Pleistocene	Futagoyama volcanics	B5	1	55/61	Cliff	N
Pliocene	Matsuura basalt	B4–B4b	2	69/72	Debris?	scattered
,,	,,	B4–B4a	3	29/31	Small cliff decomposed	N
,,	,,	,,	4	32/33	Small cliff decomposed	(R)
,,	,,	,,	5	34/38	Quarry	O(R)
,,	,,	B3–B	6	25/28	Quarry	(R)
,,	,,	,,	7	62/65	Small cliff	N
,,	,,	B2–B2d	8	01/06	Debris?	scattered
,,	,,	B2–B2c	9	07/12	Cliff	N
,,	,,	B2–B2b	10	13/18	Quarry	(R)
,,	,,	,,	11	19/24	Quarry	O(R)
,,	,,	,,	12	45/47	Quarry	O(R)
,,	,,	,,	13	66/68	Cliff	(R)
,,	,,	B1–B1d	14	91/95	Outcrop poor decomposed	N
,,	,,	B1–B1c	15	73/78	Small cliff	(R)
,,	,,	B1–B1b	16	79/81	Outcrop poor decomposed	very weak
,,	,,	B1–B1a	17	39/44	Old quarry	R
,,	,,	,,	18	48/54	Quarry	N
Upper Miocene	Sanukitoids	S	19	82/85	Quarry	R
,,	Yamazakiyama trachytic andesite	TA	20	86/90	Quarry	O(R)

† Ref.: Explanatory texts for Geological Map of Japan, ' Yobuko ' and ' Karatsu ' (Kobayashi, Inai & Matsui 1955–6) and ' Imari ' (Imai, Sawamura & Yoshida 1958).
* measured in this study, see Section 2.3.

and site numbers are listed in Table 1. Most of the samples were collected from the Matsuura Basalt which consist in this area of eleven lava 'Flow Groups' called B1 (a, b, c, d): B2 (a, b, c, d): B3 and B4 (a, b), The Matsuura basalts are overlain by the Pleistocene Futagoyama volcanics (B5) and overlie the Miocene Sanukitoids (S) and the Yamazakiyama trachytic andesite (TA). All these formations lie almost horizontally, forming a plateau, and no correction was made for their small tilt. A map showing the location of the sites is presented in Fig. 1.

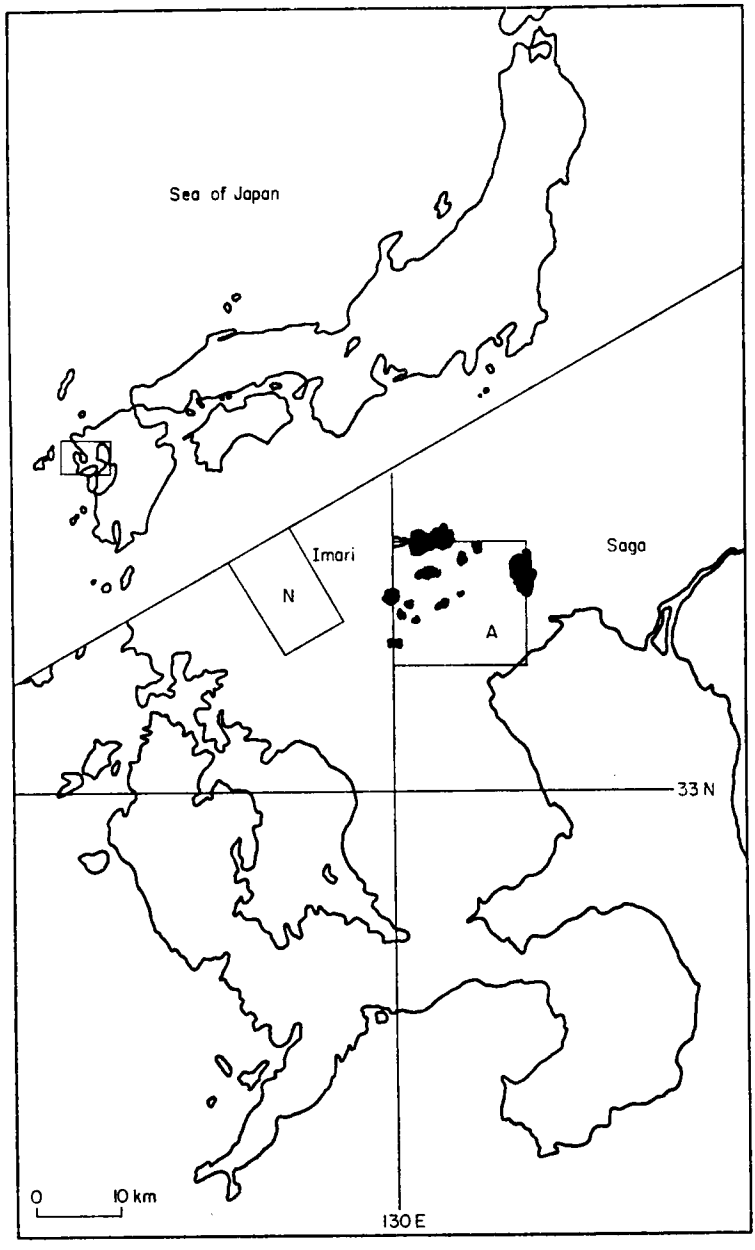


FIG. 1. Map of sampling area. The area from which our samples were collected is labelled 'A'; that from which Nomura's were collected is labelled 'N'.

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Table 2

Site-mean directions of cleaned R.M.

Flow name	Site	N	D	I	α	k	δ
B5	1	7	8°	+44°	3°	833	3°
B4b	2	4			scattered		
B4a	3	3	215°	−15°	13°	81	9°
	4	2	244°	+17°	9°	816	3°
	5	5	9°	+36°	5°	206	5°
B3	6	3	220°	−24°	18°	48	11°
	7	4	2°	+44°	3°	787	3°
B2d	8	5			unstable		
B2c	9	6	2°	+41°	3°	326	4°
B2b	10, 12	9	162°	+35°	5°	108	8°
	11	6			unstable		
	13	3	235°	−67°	7°	271	5°
B1d	14	5	moves towards normal axial dipole field direction on cleaning				
B1c	15	6	204°	−55°	8°	68	10°
B1b	16	3			very weak		
B1a	17	6	179°	−47°	6°	132	7°
	18	6	1°	+30°	5°	138	7°
S	19	4	198°	−48°	8°	122	7°
TA	20	5	117°	−30°	18°	18	19°

N = number of samples.
D, I = declination and inclination of cleaned remanence.
α = radius of 95% cone of confidence.
k = precision.
δ = circular standard deviation.

At least two to three specimens were drilled in the laboratory from each hand sample. The N.R.M. was measured and alternating field demagnetization was carried out. First, pilot specimens from each hand sample were demagnetized progressively in eight stages, usually up to 630 Oersteds* but occasionally up to 850 Oersteds. Field strengths for demagnetizing the bulk of the specimens were then selected. The directions of cleaned R.M. so obtained were averaged stage by stage, sample-mean directions and then site-mean directions (Table 2) being calculated. The stability of the remanent magnetization measured within each site is briefly described in an appendix.

2.2 *Normal, reversed and oblique remanent magnetization*

The directions of N.R.M. and of cleaned R.M. of all the specimens studied are illustrated in Figs 2*a* and *b*. It will be noticed that the cleaned R.M. of a considerable proportion of the specimens makes large angles with the axial dipole field direction and cannot simply be classified as normal or reversed. We present Figs 3, 4 and 5 to illustrate the stability of these obliquely magnetized specimens. Fig. 3 illustrates the result that specimens from site 17 are all reversely magnetized while those from site 18 are all normally magnetized. The A.F. coercivity of specimens from site 18 is slightly higher than that at site 17. We will not follow up this observation now, because the point we wish to make is that the A.F. coercivities of obliquely magnetized samples from sites 10, 11, 12 and 13 fall in the same range as illustrated by Fig. 4. The R.M. directions of the obliquely magnetized specimens from site 11 are more scattered and the A.F. coercivities of many of these are lower than those of the specimens from sites, 10, 12 and 13 whose R.M. directions show tight grouping.

* Peak field values are quoted throughout.

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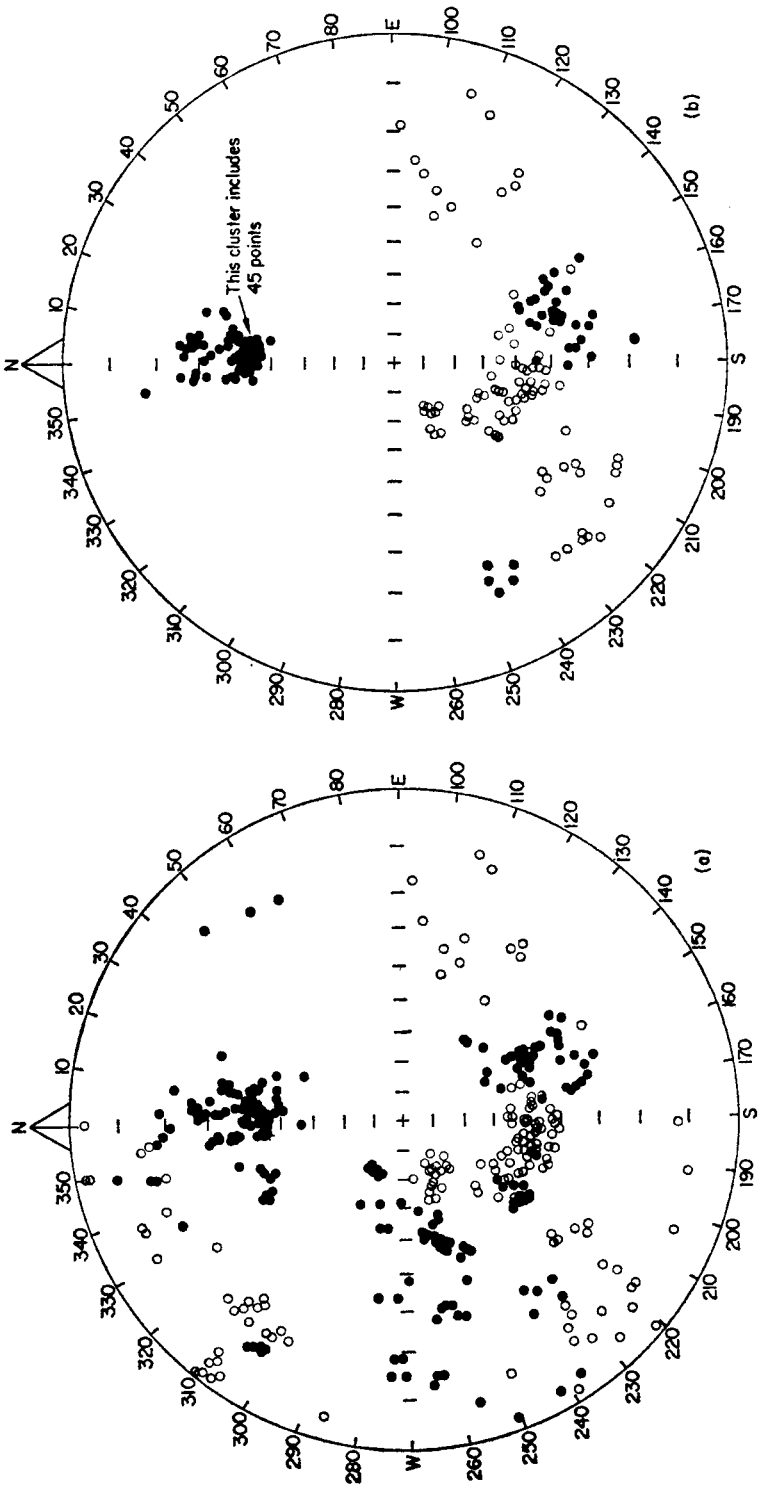


FIG. 2. Specimen-mean directions (*a*) before and (*b*) after cleaning.

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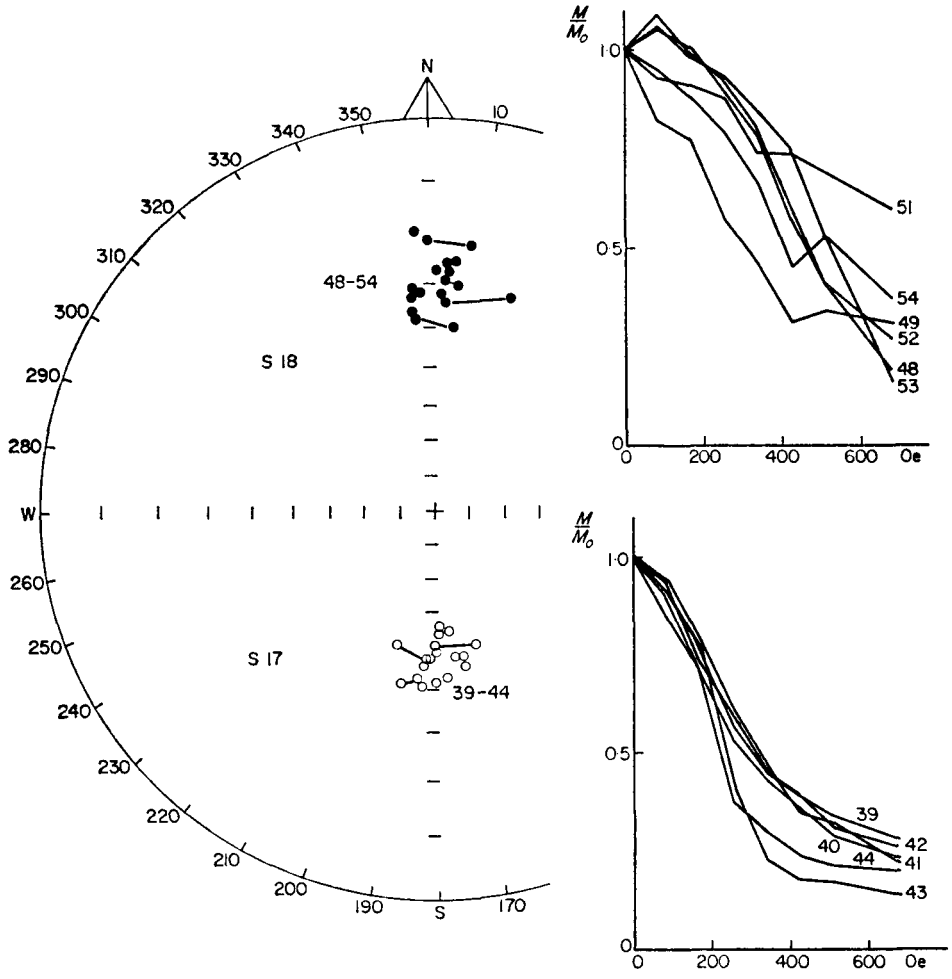


FIG. 3. Examples of N.R.M. directions and A.F. demagnetization curves for stable normal and reversed samples (Flow Bla).

Characteristic behaviour of specimens with low A.F. coercivities is illustrated by Fig. 5. The N.R.M. directions at site 1 are oblique but scattered. They exhibit normal polarity after cleaning, sequences of directions of R.M. at successive stages of cleaning being illustrated in Fig. 5*c*.

2.3 *Virtual geomagnetic poles*

Virtual geomagnetic poles computed from the cleaned R.M. of each sample are illustrated in Fig. 6 which shows that while the poles situated in the Northern Hemisphere are tightly grouped, those situated in the Southern Hemisphere are scattered. Site-mean virtual poles are listed in Table 3. We have classified the reversely magnetized sites into three sub-groups which we call R, (R) and O(R). Sites belonging to sub-group R have truely reversed polarity, i.e. their virtual poles are comparatively close to the geographic pole, within 25° of it. The virtual poles of sub-group (R) lie between 25° and 45° of the geographic pole while those of sub-group O(R,) (meaning obliquely magnetized but closer to reversed than normal) lie between 0° and 45° of the geographic equator. All five normally magnetized sites

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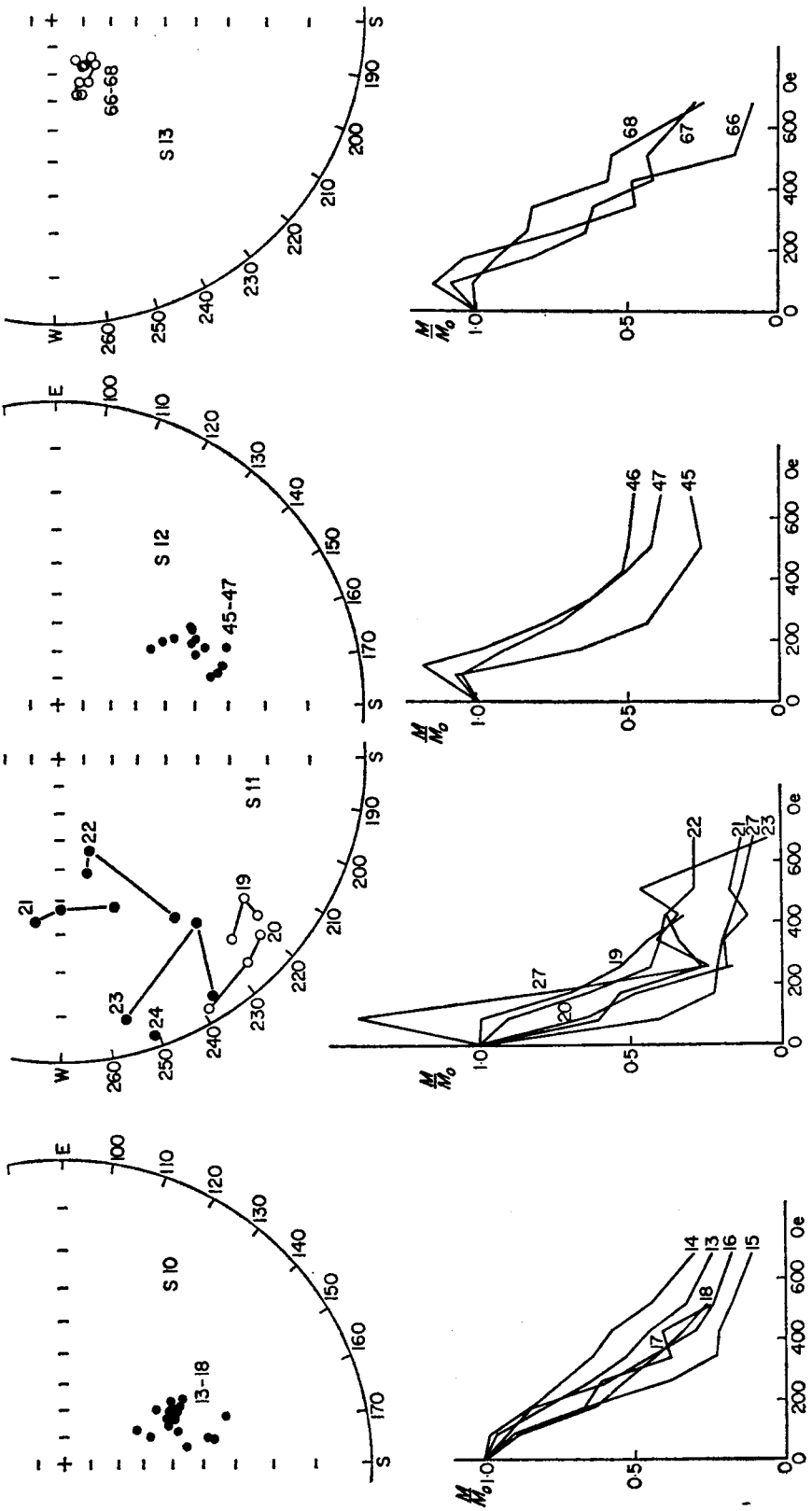


FIG. 4. Examples of N.R.M. directions, A.F. demagnetization curves for samples with oblique magnetization (sites 10, 11, 12 and 13 from Flow B2b).

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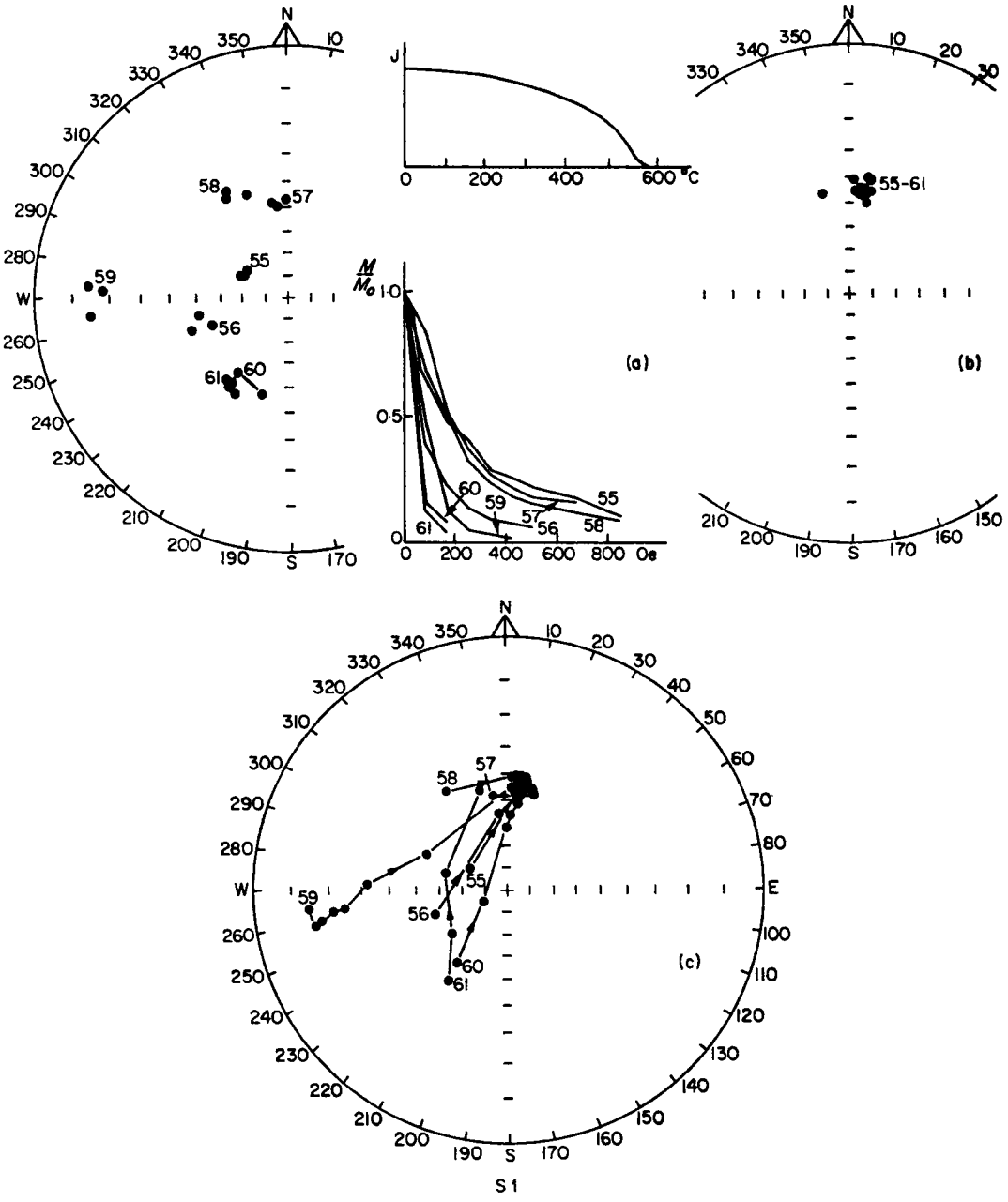


FIG. 5. (*a*) Examples of N.R.M. directions with A.F. demagnetization curves and a T.M. curve for samples having soft components of magnetization (Flow B5). (*b*) Cleaned R.M. directions and (*c*) changes in direction of R.M. during cleaning.

belong to sub-group N: sites belonging to sub-groups (N) and O(N), i.e. the normal equivalents of (R) and O(R), were not found in our collection.

We have seen that the high between-site scatter of reversely magnetized sites is not due to relative instability or within-site scatter of reversely magnetized samples. The distribution of these virtual poles is not random: they are mainly situated between meridians 10° W and 80° E (Fig. 6*b*) and suggesting that these lavas were erupted

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FIG. 6. Sample-mean virtual pole positions for stable hand samples.

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Table 3

Palaeomagnetic North Pole for each site

Site	Lat.	Long. (°E)	α	k	δ	Polarity class
1	79° N	262°	3°	1230	3°	N
2		scattered				
3	48° S	69°	14°	80	9°	(R)
4	16° S	62°	4°	3034	2°	O(R)
5	74° N	276°	5°	190	6°	N
6	48° S	59°	15°	65	10°	(R)
7	82° N	296°	3°	677	3°	N
8		unstable				
9	80° N	297°	4°	255	5°	N
10 & 12	34° S	150°	5°	99	8°	O(R)
11		unstable				
13	46° S	359°	10°	134	7°	(R)
14		moves towards normal axial field direction on cleaning				
15	69° S	22°	10°	48	11°	(R)
16			very weak			
17	85° S	138°	6°	105	8°	R
18	73° N	305°	4°	224	5°	N
19	73° S	48°	8°	125	7°	R
20	31° S	219°	18°	18	19°	O(R)

Flow names and numbers of samples are as given in Table 2

during times of transition of polarity of the geomagnetic field. We shall discuss this further in Section 4.3.

2.4 *Palaeosecular variation*

We may treat each site-mean direction of R.M. as a fossilized record of the geomagnetic field at an instant of geological time. A number of such site-mean directions thus gives a sample of the dispersion in time of geomagnetic field directions at a given place, i.e. of the palaeosecular variation of the geomagnetic field (Creer 1962).

The circular standard deviation of the population of site-mean directions for the five sites with normal R.M. is given in entry 1 of Table 4. Since the R.M. of only two sites fall in polarity sub-class R, we have combined sub-classes R and (R) to form entry 2. Entry 3 lists the mean direction and its relevant statistics for sub-classes R+(R)+O(R) taken together.

These ' raw ' circular standard deviations, δ_t, contain a contribution due to the uncertainty with which the position of each site-mean direction has been determined. If δ_w is the C.S.D. due to the scatter of sample-mean directions within each site and δ_b the true between site C.S.D. which we seek to determine, then

$$\delta_b^2 = \delta_t^2 - \delta_w^2/N$$

where N is the number of samples per site. In fact, the within-site C.S.D. varies from site to site as does the number of samples taken per site. So we make the approximation of using the R.M.S. value of δ_w for each site and the average number, N, of samples per site in the above equation. This is how the C.S.D. values given in Table 5 were computed.

If we are to compare these palaeosecular variation (P.S.V.) dispersions with those from other parts of the world, we must make a correction for the latitude of the site. Values of between-site C.S.D. corrected to the equator are listed in Table 5. From the low P.S.V. dispersion of 8° obtained for our normally magnetized group

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of sites, we conclude that only a very short time can be represented by these: the world-wide average value is 18°±4°, Creer & Sanver (1968). The P.S.V. dispersion of 30° given by the R+(R) sub-groups of sites is on the other hand rather greater than the world-wide average value of 21°±6° for successions of flows with reversed polarity (largely from the Matuyama epoch).

These differences, however, are almost certainly because we have too few sites in our collection, which when combined with that of Nomura (1967), allows us to calculate the data presented in the middle set of entries in Table 5. The between-site C.S.D., corrected to the equator, of 19° and 25° for normal and reversed magnetizations respectively, agree very well with the world-wide averages presented at the bottom of Table 5.

3. Correlation of magnetic and mineralogical properties

3.1 *Magnetic properties averaged site by site*

A summary of the site-mean values of intensity of R.M. before and after cleaning, M_0 and M respectively, and susceptibility, k are listed in Table 6. A considerable range in the tabulated values will be noted: thus M_0 varies by a factor of at least 230 from 115 mG (for site 8) to 0·5 mG (for site 20) while M_0 at site 16 was even less then this. The range in values of susceptibility is considerably smaller, the highest value measured, viz. 7·5 mG Oe^{-1} at site 4 being only about 20 times larger than the value of 0·4 mG Oe^{-1} measured for sites 15 and 18. The mean susceptibility is less than the latter only at site 16, and omitting this site we find that the range in N.R.M. intensities is over 10 times as broad as that of the corresponding susceptibility values. If we make the generalization that susceptibility (those recorded in Table 6 are mean values for the several samples covering a range of oxidation states found within each site) is proportional to the amount of magnetic material in the rock, supposing this is on average uniform, we may deduce that the larger range of N.R.M. intensities may be accounted for by actual variations in intensity of the palaeogeomagnetic field by a factor of the order of 10.

In the following sub-sections we investigate the relationships between the various measured quantities by discussing individual samples.

3.2 *High temperature oxidation*

The extent and degree of exsolution of ' titanomagnetite ' grains may be described by assigning an oxidation stage (Wilson & Watkins 1967). It is known that this kind of exsolution must have occurred at moderately high temperatures, exceeding 600 °C (e.g. see Lindsley 1965). Heating times of 10–20 hours in the laboratory at 1000 °C in 10^{-1} torr are required to produce lamellae with dimensions of 10 microns or so

Table 4

Mean R.M. directions and virtual geomagnetic poles of polarity groups using sites as statistical units

	Polarity	Number of sites	Mean direction of R.M. D	I	α	δ	North Palaeomagnetic Pole Lat.	Long.	α	δ
1	N	5	4°	39°	6°	6°	78° N	288° E	5°	5°
2	R & (R)	6	208°	−44°	20°	23°	65° S	43° E	19°	22°
3	R, (R) & O(R)	10	194°	−25°	36°	50°	69° S	71° E	30°	42°

Notes: (i) Line 1 comprises sites 1, 5, 7, 9, 18; line 2 comprises sites 3, 6, 13, 15, 17, 19; and line 3 comprises 3, 4, 6, 10, 12, 13, 15, 17, 19, 20.
(ii) For polarity classification see Section 2.3.

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Table 5

Dispersions due to palaeosecular variations

Source of data		Polarity	Number of sites	δ_t	δ_w	δ_b	δ_0
This paper	1	N	5	6°·5	7°·0	6°·0	8°·0
	3	R & (R)	6	23°·0	37°·5	22°·0	30°·0
	4	R, (R) & O(R)	10	50°·0	65°·0	49°·5	68°·0
This paper combined with Nomura (1967)	1	N+(N)	12	14°·5	12°·0	14°·0	19°·0
	2	N+(N)+O(N)	13	18°·0	12°·0	17°·5	24°·0
	3	R+(R)	14	19°·5	23°·0	18°·5	25°·5
	4	R+(R)+O(R)	17	45°·5	31°·5	44°·5	61°·5
World average*		Normal					18°·2±4°·3
,, ,,		Reversed					20°·6±6°·0

δ_t = total C.S.D. of site-mean directions
δ_w = average within-site C.S.D.
δ_b = calculated between site C.S.D.
δ_0 = between-site C.S.D. corrected to equator

* see Creer & Sanver (1968) for details of computation.

in basalts initially containing homogeneous titanomagnetites, Petersen & Creer (1970). The oxidation stage of the different specimens studied are listed in Table 7.

3.3 *Low temperature oxidation*

Oxidation has occurred in many basalts at much lower temperatures. For instance Ade-Hall *et al.* (1968) observed a ' granular ' appearance of the titanomagnetite grains in polished sections of some basalts observed under the ore micro-

Table 6

Intensity of NRM, of R.M. and of susceptibility

Flow	Site	M_0	M	k	M/M_0	M_0/k
B5	1	4·0	2·0	3·0	0·45	1·40
B4b	2	21·0	10·5	1·0	0·50	20·30
B4a	3	1·5	1·0	6·0	0·75	0·25
	4	2·0	1·5	7·5	0·75	0·25
	5	1·5	1·0	2·5	0·60	0·50
B3	6	8·0	8·0	6·5	1·00	1·25
	7	4·0	2·0	6·5	0·55	0·60
B2d	8	115·0	—	2·5	—	44·25
B2c	9	3·0	2·0	1·0	0·60	2·60
B2b	10, 12	1·0	1·0	1·5	0·80	0·80
	11	1·5	—	6·0	—	0·25
	13	1·0	1·0	2·0	0·90	0·55
B1d	14	7·0	0·5	1·5	0·10	5·10
B1c	15	2·0	2·5	0·5	1·15	5·80
B1b	16	—	—	—	—	—
B1a	17	14·5	10·5	6·0	0·70	2·40
	18	3·0	3·0	0·5	0·90	4·50
S	19	2·0	1·0	2·0	0·40	1·10
TA	20	0·5	—	6·5	—	0·10

M_0 = G.M. of intensity of N.R.M. in milligauss (mG)
M = G.M. of intensity of cleaned R.M. in mG
k = G.M. of susceptibility in mG/Oe

Values of above given to nearest 0·5 (0·05 in last two columns).

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Table 7

Various measured physical parameters as a function of oxidation stage

Oxidation stage	Flow No.	Sample No.	Grain size (microns)	$J_s(T)$ Curve type	Curie points (°C)	M_0	k	M_0/k
1	B1a	49	40	1	270°, —	2·0	1·5	1·15
1	B1a	54	20	1	275°, —	2·5	0·5	4·35
1–2	B2d	1	25	2	320°, 440°	95·5	2·0	44·0
1–2	B2d	6	40	1–2	312°, 525°	174·5	3·0	61·0
2	B2c	8	25	4	—, 580°	2·0	1·0	4·35
4	B2c	11	25	2	400°, 570°	3·5	1·0	3·40
2	B1d	93	145	2	235°, 590°	4·5	1·5	3·20
2–3	B1d	91	145	2	300°, 570°	6·5	1·5	4·85
3	B4a	29	85	4	—, 590°	1·5	6·5	0·25
3	B4a	31	50	4	—, 590°	1·5	5·5	0·30
3	B4a	32	55	4	—, 595°	1·5	6·0	0·25
3	B4a	33	75	4	—, 590°	2·0	8·0	0·25
4	B1a	39	155	4	—, 580°	6·5	3·0	1·95
4	B1a	44	145	4	—, 585°	7·5	5·0	1·60
2	B2b	45	40	4	—, 575°	1·0	3·0	0·40
4	B2b	47	80	4	—, 580°	1·5	4·0	0·30
4	TA*	86	250	4	—, 550°	0·5	6·0	0·10
4	TA*	90	115	4	—, 550°	0·5	4·5	0·05

Notes: M_0 in milligauss and k in mG/Oe.
* is trachitic andesite
Values of above given to nearest 0·5 or 0·05.

scope and attributed this to burial at considerable depths several million years after the formation of the flow. Granulation is thus possibly a type of low temperature oxidation. Creer & Petersen (1969) oxidized homogeneous titanomagnetites contained in basalts by heating in air in the laboratory for extended times up to 1000 hours at 400 °C and noticed that the shape of the thermomagnetic curves evolved in a characteristic way with time of heating, i.e. with extent of oxidation, as shown in Fig. 7. The initial curves (which we refer to as type 1) have a single low Curie point from which, assuming a stoichiometric titanomagnetite, we may estimate the percentage of ulvöspinel which we may compare with that measured by X-ray microanalysis. A new magnetic phase gradually develops having a Curie point of just over 500 °C, the thermomagnetic curve evolving through types 2 and 3 to type 4 when the initial magnetic phase has been completely altered. The ' titanomagnetites ' in basalts oxidized in this way may sometimes appear identical to the unoxidized titanomagnetites in the untreated samples: sometimes neither granulation nor exsolution are visible under the ore microscope.

Such ' titanomagnetites ' would be placed in oxidation stage 1, yet the presence either of two Curie points or a single high Curie point gives them away. Oxidation stage 1 thus covers a wide range of degrees of low temperature oxidation.

3.4 *Oxidation stage and Curie temperature*

In Table 7 a correlation between Curie temperature and oxidation stage is seen. Both samples studied with oxidation stage 1 have a single low Curie point, with thermomagnetic curves of type 1. Samples at oxidation stage from 1–2 to 2–3 have two Curie points with type 2 thermomagnetic curves, the low Curie point being accounted for by incompletely exsolved grains and the high Curie point by exsolved regions (possibly too small to be observed under the microscope). More highly oxidized samples have a single high Curie point with thermomagnetic curves of type 4. There appears to be no obvious correlation between oxidation stage and M_0, k or M_0/k.

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3.5 *Magnetic polarity, oxidation stage and A.F. coercivity*

Next we have investigated whether various magnetic and mineralogical properties depend on the polarity of R.M. Samples have been grouped into three classes, viz. (i) of reversed polarity, (ii) of normal polarity, and (iii) with directions of R.M. so scattered they cannot reliably be placed in either class (i) or (ii). Class (i) includes all three sub-classes of reversed polarity, R, (R) and O(R) defined in Section 2.3. The number of samples having various properties are shown in Table 8.

On the whole, normally magnetized samples belong to oxidation stages 1 or 2 and thus tend to have thermomagnetic curves of type 1 or 2. They also tend to have A.F. coercivities* less than 200 Oe and to be of smaller grain size (less than 40 microns). The unstable samples tend to have low A.F. coercivities as would be expected.

On the other hand the reversely magnetized samples tend to belong to oxidation stages 3 or 4 and thus have type 3 or 4 thermomagnetic curves; they tend to have higher A.F. coercivities (200–700 Oe) and to contain larger ' titanomagnetite ' ore grains.

The higher A.F. coercivities of the reversely magnetized samples is probably associated with a higher degree of exsolution: the distribution of grain size between the two polarities is irrelevant if the grains contain exsolution lamellae, since the dimensions of these will affect the magnetic properties rather than the size of the whole grain.

It must be stressed, however, that there are insufficient data in Table 8 to rule out the possibility that the variation of properties between samples with normal and reversed polarity is not due to a biased collection. Thus, while the normal flows tend to have smaller grain size than the reversed flows, they also tend to be younger. In volcanic terrain, random sampling of flows will yield young material biased towards flow margins and older material with a greater proportion of flow centres.

There appears to be no correlation of polarity or stability with M_0 or M_0/k (see lowest group of entries in Table 8).

3.6 *Microprobe analyses*

The ' titanomagnetites ' in specimens from six samples were analysed with the electron microprobe. In the base of exsolved grains, the electron beam was scanned

Table 8

General classification of mean sample directions

Measured property	Number of samples: Reversed	Normal	Unstable or scattered	Comments
Oxidation Stages 1–2	—	4	3	Principally normal
$J_s(T)$ curve type 1–2	1	14	6	,, ,,
Low H_ca (0–200 Oe)	4	22	12	,, ,,
Small grain size (0–40μ)	1	4	1	,, ,,
Oxidation Stages 3–4	10	—	1	Principally reversed
$J_s(T)$ curve type 3–4	25	7	6	,, ,,
High H_ca (200–700 Oe)	37	6	8	,, ,,
Big grain size (40–250μ)	9	—	3	,, ,,
High M_0 (2–150 milligauss)	15	16	14	No tendency to be N or R
Low M_0 (0–2 milligauss)	26	12	6	,, ,,
High M_0/k (2–60)	7	11	14	,, ,,
Low M_0/k (0–2)	34	17	6	,, ,,

H_ca: Coercive force for $M/M_0 = 0{\cdot}5$ on the demagnetization curve.
M_0: G.M. of intensity of N.R.M. in mG.
k: G.M. susceptibility in mG/Oe.

* Defined here as peak value of demagnetizing field required to reduce the component of remanence to half its initial value.

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over an area large enough to cover several lamellae so as to obtain an average composition (Creer & Ibbetson 1970). The results and the analyses are given in Table 9. 'Compositional' Curie points, calculated for stoichiometric (unoxidized) pure titanomagnetites with the same Ti/Fe ratios as the analysed grains are also given. One would expect the measured Curie temperatures to be lower than these 'compositional' ones due to the presence of impurity trace elements (Fröhlich, Loffler & Stiller 1965) if the grains were unoxidized. However, in all the samples analysed, some exsolution has occurred and it is the exsolved iron-rich ferrimagnetic which determines the shape of the thermomagnetic curve. Unfortunately no samples belonging to oxidation class 1 were analysed, but it is often found that good agreement exists between 'compositional' and measured Curie points in such grains (Creer & Ibbetson 1970).

The average amount of titanium in the ore grains analysed, ranging from about 9 per cent to 15 per cent, is typical of that measured in other basalts studied (Creer & Ibbetson 1970).

4. Comparison with previous work

4.1 *Anomalous direction of R.M. of Sites* 10 *and* 12

The direction of remanence of these sites have S.E. declinations coupled with positive inclinations (Table 3). The magnetic vectors of samples are very tightly grouped (C.S.D. = 8°) and they are stable to A.F. treatment. Nevertheless they are strongly oblique to either the normal or reversed direction of the axial dipole field. The virtual geomagnetic poles lie in S.E. Australia (Figs 6 and 8).

At first we wondered whether an error of 180° had been made in the orientation of samples from these sites (we thought the wrong end of the compass needle might have been read). If this were so, these sites would belong to polarity class N. But we are inclined to the view that this is not the correct explanation and that the result from these two sites should be taken at their face value for the following reason.

Nomura (1967) found a similar direction of cleaned R.M. for the Nosa locality in Kyushu. The rocks there are classified as flow B2–3 (see his Table 3) while those

Table 9

Electron microprobe analyses

Sample No.		J10	J20	J30	J40	J86	J93
No. grains analysed		4	4	3	4	4	4
Degree of oxidation		2	2	2	5	3	2
Element (wt %)	Fe	52·6	48·3	54·1	53·4	55·1	47·0
	Ti	8·8	14·5	10·2	11·7	11·4	15·6
	Mg	0·7	0·9	0·6	0·7	0·3	0·9
	Al	0·8	0·9	0·9	0·7	0·3	0·8
	Cr	0·2	0·5	0·7	0·1	0·0	0·2
	Mn	0·3	0·4	0·3	0·3	0·7	0·4
	Ca	0·5	0·1	0·2	0·1	0·2	0·4
	P	1·5	0·4	0·3	0·4	0·5	0·4
	Si	1·0	0·7	0·3	0·2	0·2	0·2
	F	1·01	0·97	1·02	1·05	1·07	1·03
	A	56·6	51·5	56·6	55·9	57·4	50·3
	B	10·5	15·6	10·8	12·1	11·7	15·9
	B/A	0·186	0·304	0·191	0·216	0·203	0·316
Composition, x		0·53	0·79	0·55	0·60	0·58	0·81
Compositional T_c		185 °C	0 °C	175 °C	135 °C	150 °C	−15 °C
Measured T_c		—, 590 °C	—, 565 °C	—, 570 °C	—, 585 °C	—, 550 °C	235, 590 °C

1. Degree of oxidation is as defined by Wilson & Watkins (1967).
2. x is the parameter in the formula $(1-x)\,Fe_3O_4\,x\,Fe_2TiO_4$.
3. F is a normalizing factor.
4. A is the wt % of elements except Ti and Si expressed as Fe.
5. B is the wt % of Ti and Si expressed as Ti.

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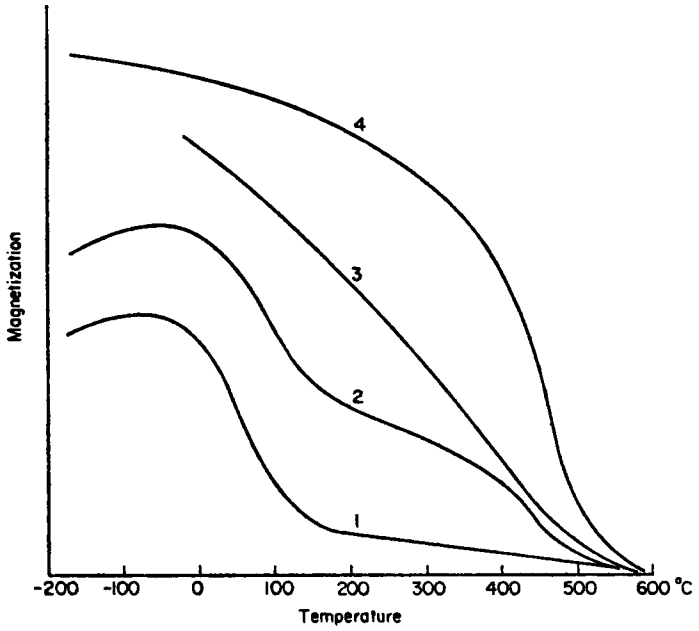


FIG. 7. Defining thermomagnetic curves of types 1, 2, 3 and 4.

from our sites 10 and 12 belong to flow B2. Thus it is quite possible that Nomura's rocks from Nosa and our sites 10 and 12 are contemporaneous. The mean direction of R.M. for the Nosa locality is given as $D = 220°$, $I = +58°$, $\alpha = 12°$ with a virtual pole at 9° S, 100° E. We also note that similar N.R.M. directions were found by Kawai *et al.* (1962) where they are illustrated in their Fig. 1.

It is thus possible that oblique directions of R.M. such as this may prove to be useful stratigraphic markers since it is unlikely that they are repeated exactly or often in the geomagnetic record like directions close to the axial dipole field.

4.2 *On the absence of polarity sub-classes (N) and O(N)*

This must be just a matter of chance in our collection because Nomura (1967) did find flows which can be so classified. The virtual poles corresponding to sites with such polarities are illustrated in Fig. 8, in which our results are plotted together with Nomura's.

4.3 *A sequence or virtual geomagnetic poles during a polarity reversal of the main field*

In the Introduction we mentioned that Nomura (1967) recorded the presence of virtual poles in low latitudes which he thought might correspond to geomagnetic field directions during a transition in polarity of the main field. Curiously, nearly all of these lie in Russia and the Indian Ocean. In Fig. 8a we show how certain of our virtual poles which we have classified as (R) or O(R) allow us to complete a sequence running from the Northern into the Southern Hemisphere, a sequence which may define the successive geomagnetic pole positions during one particular main field reversal. Many of these poles are computed from R.M. directions of Flows 4 but also the same path can be drawn between virtual pole positions from Flow 1 which is much older. This result is discussed together with other reported anomalous virtual geomagnetic pole positions derived from other formations by Creer & İspir (1970).

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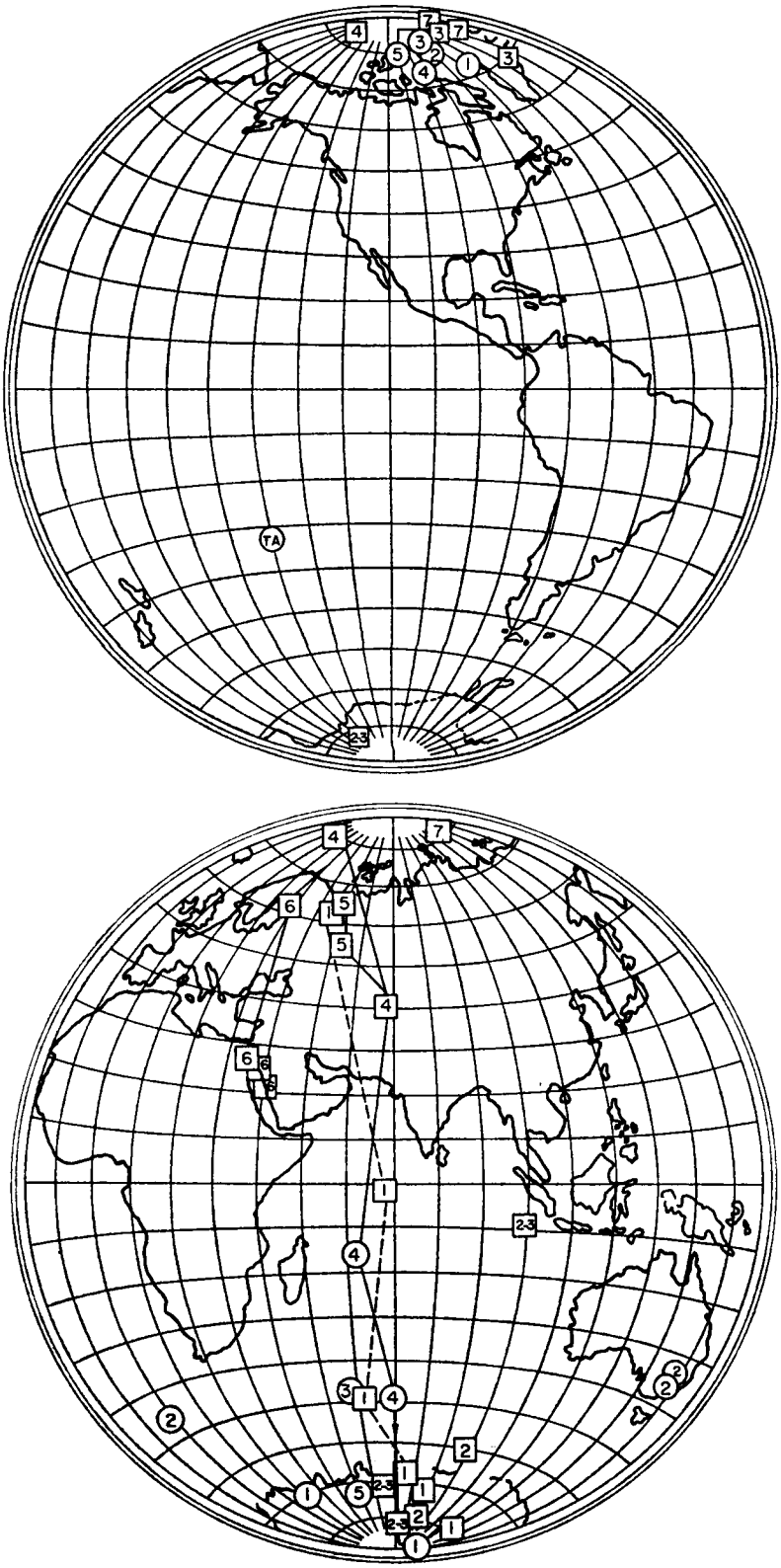


FIG. 8. Virtual poles computed for our results (circles) and Nomura's results (squares). The flow numbers are shown. Two paths of the virtual pole during reversals are shown (*a*) for Flows B4 and 5 and (*b*) for Flow B1.

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4.4 *On the intensity of the geomagnetic field during a polarity reversal*

Nomura (1967) investigated the relationship between J_n/J_{tc} (the ratio between the intensity of N.R.M. to that of the thermally cleaned R.M.) and the latitude of the virtual geomagnetic pole (see his Figs 13 and 14). He reported a decrease in this ratio for virtual poles near the present equator for the Matsuura basalts from N.W. Kyushu. Momose (1963) had previously made a similar observation.

But this ratio may depend on the state of exsolution of the ' titanomagnetites ' contained in the basalts rather than on the intensity of the geomagnetic field in which the rocks acquired their T.R.M. so we have preferred to use the ratio M_0/k, i.e. of the N.R.M. intensity to the low field reversible susceptibility. The former may be regarded as a rough measure of the ancient geomagnetic field strength and the latter to be roughly proportional to the amount of ferrimagnetic mineral present.

We have plotted the ratio M_0/k against the latitude of the virtual geomagnetic pole for the samples we have studied in Fig. 9 where its tendency may be observed for low virtual pole latitudes to be correlated with low M_0/k ratios, i.e. with low geomagnetic field intensities.

We have already noted in Section 3 that the ration M_0/k cannot be correlated with any other rock magnetic parameter.

4.5 *Correlation between magnetic polarity and the stratigraphy*

We now enquire about the extent to which the magnetic polarity determined for our collection agrees with that determined by Nomura (1967) for his collection. The two collecting sites are about 10 km apart.

We conclude that each ' Flow Group ' as defined stratigraphically (Section 2.1) must include a number of lava flows which were extruded over a period of time (Table 10). In particular Flow Group B1 contains all sub-groups of geomagnetic polarity when our results and Nomura's are considered together.

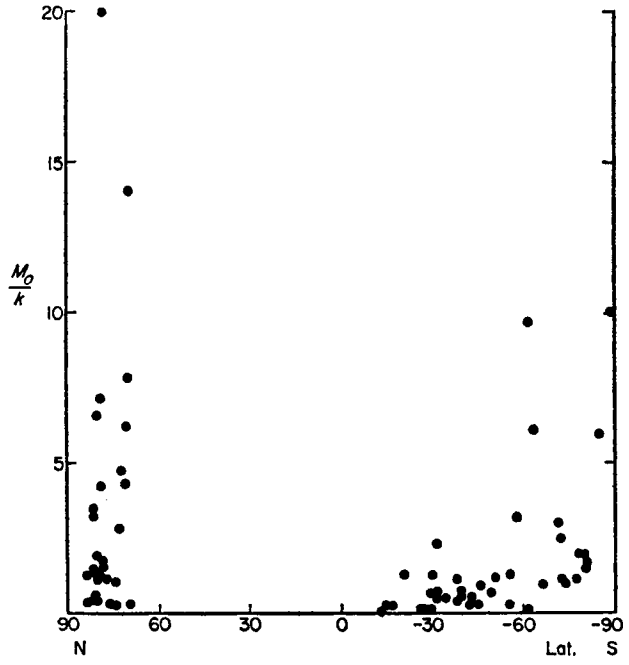


FIG. 9. The ratio M_0/k is plotted against the latitude of the virtual geomagnetic pole for each sample studied.

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Table 10

Comparison of polarities determined by this work and by Nomura's

Flow	Our polarity	Nomura's polarity	Agreement
B5	N	(N)	+
B4	N, (R), O(R)	N, N, O(N)	+−−
B3	(R), N	N, N	−+
B2–3	—	R, R, O(R)	
B2	N, (R), O(R), O(R)	R, R, (R)	−++
B1–2	—	O(R)	
B1	N, (R), R, N	(N), O(R), (R), R, R, R	+++−
S	R	—	
B_0	—	R, R	
TA	O(R)	—	

N.B. 1. The two sampling sites are about 10 km apart and are illustrated in Fig. 1.
2. The polarity symbols have the following meanings: (i) N, R: palaeomagnetic poles are situated in latitudes higher than 65°; (ii) (N), (R): at latitudes between 65° and 45°; and (iii) O(N), O(R), i.e. oblique normal or oblique reversed at latitudes between 45° and 0°.

The older flows, *S*, B_0 and *TA* exhibit reversed polarity exclusively. The youngest flow B5 exhibits normal polarity in both collections. Flow B4 is predominantly normal in Nomura's work and predominantly reversed in ours. But the longitude of our poles and Nomura's are approximately 90 °E and as already mentioned in Section 4.3 are thought to record a main field reversal.

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References

Ade-Hall, J. M., Khan, M. A., Dagley, P. & Wilson, R. L., 1968. A detailed opaque petrological and magnetic investigation of a single Tertiary lava flow Skye, Scotland II; Spatial variations of magnetic properties and selected relationships between magnetic and opaque petrological properties. *Geophys. J. R. astr. Soc.*, **16**, 389–399.

Creer, K. M., 1962. The dispersion of the geomagnetic field due to secular variation and its determination in ancient times from palaeomagnetic data, *J. geophys. Res.*, **67**, 1899–1906.

Creer, K. M. & Ibbetson, J. D., 1970. Electron microprobe analysis of homogeneous titanomagnetite contained in basalts (in preparation).

Creer, K. M. & Petersen, N., 1969. Thermochemical magnetization in basalts. *Z. Geophys.*, **35**, 501–516.

Downloaded from https://academic.oup.com/gji/article/20/2/127/595040 by Curtin University Library user on 26 May 2026

Creer, K. M. & Sanver, M., 1968. Secular variations of the geomagnetic field in the Quaternary and late Tertiary. Paper presented at N.A.T.O. conference ' Palaeogeophysics ', University of Newcastle upon Tyne, April, 1968. Published in *Palaeogeophysics*, ed. S. K. Runcorn, Academic Press, pp. 91–100.

Creer, K M. & İspir, Y., 1970. On the behaviour of the geomagnetic field during polarity transitions, *Phys. Earth Planet. Int.*, **2**,

Fröhlich, F., Loffler, H. & Stiller, H., 1965. Interpretation of changes in Curie-temperatures observed in rocks. *Geophys. J. R. astr. Soc.*, **9**, 411–422.

Kawai, N., Ito, H. & Kume, S., 1961. Deformation of the Japanese Islands as inferred from rock magnetism. *Geophys. J. R. astr. Soc.*, 127–130.

Kawai, N., Kume, S. & Ito, H., 1962. Study on magnetization of the Japanese rocks. *J. Geomagn. Geoelect., Kyoto*, **13**, 150–153.

Kobayshi, I., Imai, I. & Matsuri, K., 1955–6. Yobuko & Karatsu. Explanatory text of the geological map of Japan.

Lindsley, D. J. J., 1965. Annual Report of the Director of the Geophysical Laboratory, Carnegie Institution.

Momose, K., 1963. Studies on the variations of the Earth's magnetic field during Pliocene time. *Bull. Earthq. Res. Inst.*, **41**, 487–534.

Nomura, S., 1967. On the properties of the palaeomagnetic polarity of the Neogene and Quaternary volcanic rocks in Japan. *Gunma J. Liberal Arts Sci.*, **1**, 11–35.

Petersen, N. & Creer, K. M., 1970. Composition of phases produced by oxidation of homogeneous titanomagnetites. In preparation.

Wilson, R. L. & Watkins, N. D., 1967. Correlation of petrology and natural magnetic polarity in Columbia Plateau basalts. *Geophys. J. R. astr. Soc.*, **12**, 405–424.

APPENDIX

NOTES ON THE R.M. MEASURED AT EACH SITE

Site 1. *Flow B5* (*Fig.* 5).

The N.R.M. directions all had downward inclinations, but the scatter was high. It was strikingly reduced by A.F. demagnetization in 340 Oe. A.F. demagnetization of pilot specimens showed coercivities of over 200 Oe for samples 55, 57 and 58 whose N.R.M. directions lie close to the axial dipole field. The N.R.M. directions of samples 56 and 59 which have coercivities of about 150 Oe were moderately far away from the normal axial dipole field while the N.R.M. directions of samples 60 and 61, whose A.F. coercivities were the lowest (less than 100 Oe), lie far from the axial dipole field.

Site 2. *Flow B4b.*

The N.R.M. was badly scattered and only showed slight improvement after cleaning.

Sites 3, 4 *and* 5. *Flow B4a.*

Directions of N.R.M. of samples from sites 3 and 4 were scattered in the S.W. quadrant. Although this scatter was considerably reduced after A.F. treatment the directions of R.M. remained oblique, showing no tendency to move either towards the normal or the reversed direction of the axial dipole field. The N.R.M. of samples from site 5 was well grouped and remained of unchanged normal polarity after A.F. treatment.

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Sites 6 *and* 7. *Flow B*3.

The cleaned directions at both sites were nearly identical with the natural ones and the scatter was not significantly reduced after treatment. While the directions of R.M. of the samples from site 7 are close to the normal A.D.F., those for the samples from site 6 make an appreciable angle with the reversed A.D.F. The A.F. coercivities of the latter samples are higher than those of the former showing that this somewhat oblique direction is quite stable.

Site 8. *Flow B*2*d*.

The N.R.M. directions had negative inclinations and north-westerly declinations. A.F. cleaning was carried out on pilot specimens in up to 680 Oe, but most of the R.M. was destroyed in less than 100 oersteds. The directions of R.M. moved towards that of a normal axial dipole field but since the intensities became very weak it was not possible to measure the directions precisely. They are appreciably scattered. Two polished specimens (1 and 6) were examined (Table 7).

Site 9. *Flow B*2*c*.

Both the N.R.M. and the cleaned R.M. had normal polarity and the scatter was not significantly reduced by cleaning. Two polished specimens (8 and 11) were examined (Table 7).

Sites 10, 11, 12 *and* 13. *Flow B*2*b*. (*Fig.* 4).

The directions of cleaned R.M. of all samples from sites 10 and 12 remained anomalous (like the N.R.M.), having downward inclinations with southerly declinations. The N.R.M. directions of samples from site 11 were widely scattered both before and after cleaning. Samples from site 13 had R.M. with oblique directions which were closer to that of a reversed axial dipole field than those at sites 10 and 12 both before and after A.F. cleaning.

Site 14. *Flow B*1*d*.

The N.R.M. directions were widely scattered. The directions of R.M. of pilot specimens from samples 93 and 95 after A.F. cleaning in 510 and 680 Oe, showed a tendency to move towards the axial dipole field direction but they became too weak to measure after treatment in stronger fields. The directions of R.M. of none of the other samples changed after A.F. treatment and the scatter remained high. Polished sections of specimens 91 and 93 were examined under reflected light (Table 7).

Site 15. *Flow B*1*c*.

The N.R.M. directions were close to the reversed direction of the axial dipole field. The grouping was slightly improved by A.F. cleaning.

Site 16. *Flow B*1*b*.

All the rocks sampled here were too weakly magnetized to measure N.R.M. or initial susceptibility.

Sites 17 *and* 18. *Flow B*1*a*. (*Fig.* 3).

Thirteen hand samples were collected from two sites. At site 17 (samples 39–44), the N.R.M. directions were of reversed polarity while at site 18 (samples 48–54) they were of normal polarity. They did not change significantly during demagnetization. The samples with normal polarity appeared to have higher A.F. coercivities than those with reversed polarity. Four polished specimens, two of each polarity (39, 44, 49 and 54) were examined in reflected light (Table 7).

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Site 19. *Sanukitoids* (*S*).

Four hand samples were collected from a quarry. The N.R.M. directions were well grouped and their polarity was reversed with respect to the present geomagnetic field. The A.F. coercivity was about 200 Oe and the direction of R.M. did not change in response to this treatment so the N.R.M. directions were taken as representative of the ancient geomagnetic field.

Site 20. *Yamazakiyama Trachytic Andesite* (*TA*).

The N.R.M. directions made angles of between 20° and 80° with the reversed axial dipole field direction and they showed no appreciable change after cleaning in up to 680 Oe. Hence we consider that they represent a non-axial geomagnetic field. Two samples (86 and 90) from this site were polished and examined under reflected light (Table 7).

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